Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech (EE) Examination

May 2018

EE763 Modeling and Simulation of Renewable Energy Sources

Date: 08.05.2018, Tuesday **Time: 01.30 p.m. To 04.30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Explain the differences between conventional synchronous generator and WTGS. **[05]**

OR

Q - 1 State and explain briefly the local impacts and side wide impacts of renewable energy sources on power system. **[05]**

Q - 2 **Answer Any Two. (Each question carry seven marks)** **[14]**

(1) Derive the equations of 5th order FSIG model as a voltage behind a transient reactance.

(2) Write a note on mathematical modelling of any one type of static var compensators used in wind diesel system.

(3) Draw a typical maximum power point tracking (MPPT) characteristics of wind turbines. Explain its operating regions.

Q – 3 **Answer Any Two. (Each question carry eight marks)** **[16]**

(1) Explain briefly the FSIG, DFIG and FRC configurations of wind turbine generating system with neat schematic along with their merits and demerits.

(2) Explain the working principle and operation of DFIG in synchronous, super synchronous and sub synchronous mode with neat diagram.

(3) Explain the Torque Control Scheme applied to RSC of DFIG with necessary equations along with diagram.

SECTION – II

Q - 4 Draw the flow chart of hill climbing method for maximum power point tracking of solar photovoltaic system and explain the same in brief. **[07]**

Q – 5.a Explain the effect of change in temperature and insolation on the I-V curve of solar photovoltaic system. **[07]**

Q – 5.b Draw the block diagram of grid connected photovoltaic system and explain it in brief. **[07]**

OR

Q – 5.a Draw and explain I-V curve of photovoltaic system with battery as a load. **[07]**

Q – 5.b Derive the voltage and current equations from equivalent circuit of photovoltaic cell. Also write these equations at standard test condition. **[07]**

Q – 6. Answer Any Two. (Each question carry seven marks) **[14]**

(1) Give comparison of various fuel cells.

(2) Explain gas diffusion process at electrodes of SOFC and PEMFC.

(3) Write a short note on sun tracking system.

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